

$$\text{maximize } \mathbf{c}^\top \mathbf{x} \tag{1}$$

$$\text{s.t.} \qquad \mathbf{a}_i^\top \mathbf{k} \leq \mathbf{b} \qquad \forall i \in I \tag{2}$$

$$\begin{aligned} d_1d_1 + d_2x_2 \\ + d_3x_3 + d_4x_4 \leq f \end{aligned} \tag{3}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{x} = \mathbf{x} \textit{whomadethisonesolong} \tag{4}$$

$$\mathbf{x} \geq \mathbf{0} \tag{5}$$